

LaTeX 101: The Architect's Blueprint

Moving from Manual Formatting to Structural Design.

By Preethi Siva Kumar and Dr. Kartik Bulusu

MAE 1001 - Fall 2025

| The Manual Formatting Trap

The "Housekeeper" Problem

Have you ever finished a 10-page report only to realize the margins are wrong?

- Fixing headers one by one.
- Aligning images by dragging them pixel by pixel.
- Fighting with numbering that breaks automatically.

```
# The Result:  
Error: Inconsistent_Formatting_Exception  
Time_Lost: 3_Hours
```



The Factory Model

Think of **Microsoft Word** as a *Housekeeper*. You must manually fold every towel and place every pillow. If the style changes, you re-do every room.

Think of **LaTeX** as an *Automated Factory*. You simply write the blueprint (the code). The robots build the document perfectly, every single time.

Focus on the Rules, not the Labor.



| Word vs. LaTeX

Feature	Microsoft Word (Housekeeper)	LaTeX (Architect)
Method	Visual (WYSIWYG) - Drag & Drop	Code (Markup) - Logical Tags
Focus	Appearance (How it looks)	Structure (What it means)
Consistency	Hard to maintain in long docs	Guaranteed by the compiler
Analogy	Building with LEGOs manually	Writing a script to build the city

My LaTeX Projects

5 Simulation Results and Discussions

5.1 Part 1: Wing with Single Thickness

The analysis performed in Part 1, using a uniform thickness for the skin, ribs, and spars, produced the following structural and performance results. For this and subsequent analyses, the maximum and minimum pressures across the bottom surface of the wing are taken as 1333.33 Pa at the root and 333.33 Pa at the tip, respectively. The pressure distribution along the wing is modeled by the following equation:

$$p(z) = -\frac{1000z}{27} + \frac{4000}{3} \quad (3)$$

The thickness selected for Part 1 is 0.00412 meters. This value represents the maximum allowable thickness that still satisfies the overall structural weight constraint of 6000 kg. It was chosen through an iterative design process to ensure optimal use of available mass while maintaining structural integrity and meeting safety requirements.

5.1.1 Displacement Analysis

Figures 10 and 11 show the Y-direction displacement for the spars, ribs, and skin in Part 1, where all components were assigned the same thickness. Maximum displacement occurs at the wing tip, which is expected due to the cantilever configuration and pressure load.

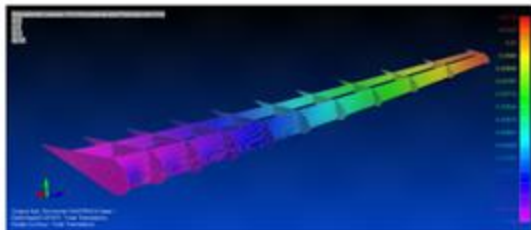


Figure 10: Total Displacement at Spars and Ribs

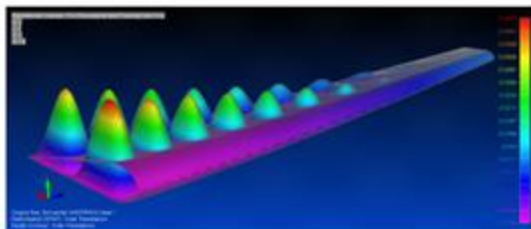


Figure 11: Total Displacement at Skin

Table 4 lists the peak displacement values in all three global directions. The Y-displacement, at 0.37 m, is the maximum displacement observed on the wing skin. Displacements in the X and Z directions are significantly lower and mainly result from torsion and bending effects.

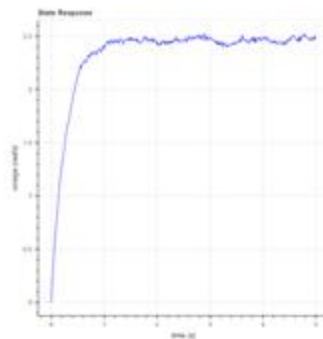


Figure 1: Bode Plot for State Response

Parameter	Description	Value	Units
R_m	Resistance of motor coil	2	Ω
J	Moment of Inertia of wheel	0.004	kg m^2
c	Linear Damping Coefficient of wheel	0.015	$\frac{\text{N s}}{\text{m}}$
K_v	Motor Constant	100	$\frac{\text{N}}{\text{A}}$
η	Motor/Gearbox efficiency	0.9	

Table 1: Model Parameters

$$J\dot{\omega} = \frac{\eta}{R_m K_v} v_{in} - \left(c + \frac{\eta}{R_m K_v^2}\right)\omega$$

$$\dot{\omega} = \frac{\eta}{R_m K_v J} v_{in} - \left(\frac{c}{J} + \frac{\eta}{R_m K_v^2 J}\right)\omega$$

Substituting the provided values and solving the equation with the RK-4 method [8],

```
import numpy as np
import matplotlib.pyplot as plt

def w_dot_func(t, w, v_in):
    w_dot = w + v_in - b * w
    return w_dot

def rk4(t, t0, w0, h, num_steps, v_in):
    solution = [(t0, w0)]
    t = t0
    w = w0

    for _ in range(num_steps):
        k1 = h * f(t, w, v_in)
        k2 = h * f(t + 0.5 * h, w + 0.5 * k1, v_in)
        k3 = h * f(t + 0.5 * h, w + 0.5 * k2, v_in)
        k4 = h * f(t + h, w + k3, v_in)
        w = w + (k1 + 2*k2 + 2*k3 + k4) / 4
        t = t + h
        solution.append((t, w))

    return solution
```

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My LaTeX Projects

```
\thispagestyle{plain}
\tableofcontents
```

```
\newpage
\renewcommand{\listtablename}{LIST OF TABLES}
\listoftables
\newpage
```

```
\thispagestyle{plain}
\renewcommand{\listfigurename}{LIST OF FIGURES}
\listoffigures
\newpage
```

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x

| The Two Parts of a Document

1. The Preamble

This is the **Blueprint**. It defines the global rules: margins, fonts, and packages.

```
\documentclass{article}  
\usepackage{geometry}  
\title{My Physics Report}
```

2. The Body

This is the **Content**. You type text and "tag" it with structure.

```
\begin{document}  
  \maketitle  
  \section{Introduction}  
  Hello World!  
\end{document}
```

The "Killer Feature": Math

Clean Code

Write complex math using simple logic.

```
x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
```

Professional Output

The compiler renders it with print-quality precision.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Exercise: The Physics and Probability of Rolling Dice

We will create a report in 20 minutes.

Your Mission:

- Set up the document structure.
- Write the probability equations.
- Add the dice image
- **DO NOT** touch the fonts or margins manually.

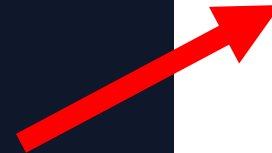


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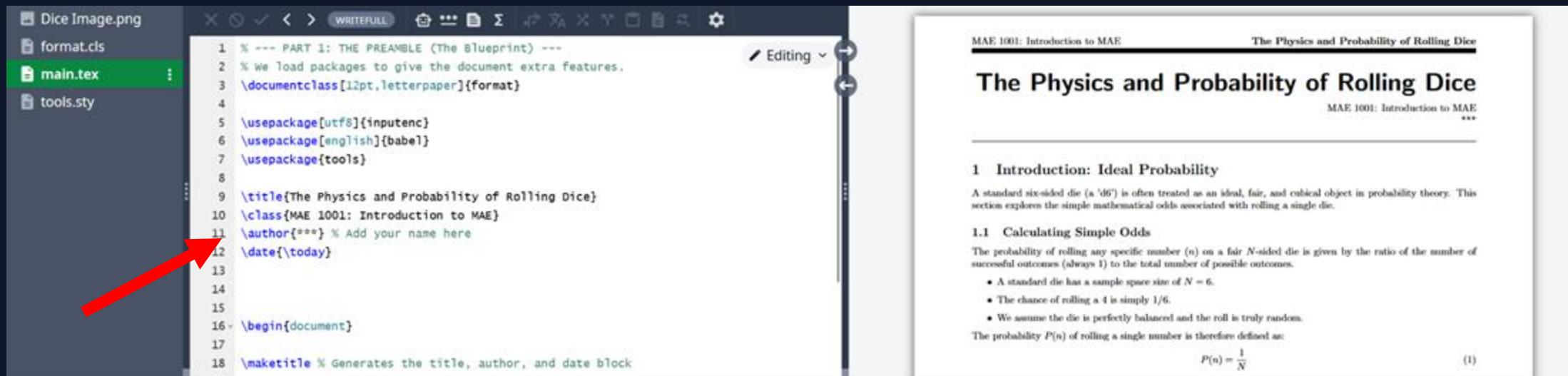
Password



Log in

The Workflow

1. Import the Zip File on Overleaf from EdStem
 - a. New Project
 - b. Upload Project
 - c. Select a ZIP
 - d. Go to main.tex
2. Add your Name
3. Recompile



The screenshot displays the Overleaf editor interface. On the left, a file explorer shows the project files: Dice Image.png, format.cls, main.tex (selected), and tools.sty. The main editor area shows the LaTeX source code for main.tex. A red arrow points to line 11, which contains the command `\author{***}` with a comment "% Add your name here". The code includes a preamble with package loading and document class settings, followed by a document body starting with `\begin{document}`. The right pane shows the compiled PDF output, which features the title "The Physics and Probability of Rolling Dice" and the author information "MAE 1001: Introduction to MAE ***".

```
1 % --- PART 1: THE PREAMBLE (The Blueprint) ---
2 % We load packages to give the document extra features.
3 \documentclass[12pt,letterpaper]{format}
4
5 \usepackage[utf8]{inputenc}
6 \usepackage[english]{babel}
7 \usepackage{tools}
8
9 \title{The Physics and Probability of Rolling Dice}
10 \class{MAE 1001: Introduction to MAE}
11 \author{***} % Add your name here
12 \date{\today}
13
14
15
16 \begin{document}
17
18 \maketitle % Generates the title, author, and date block
19
```

MAE 1001: Introduction to MAE The Physics and Probability of Rolling Dice

The Physics and Probability of Rolling Dice

MAE 1001: Introduction to MAE

1 Introduction: Ideal Probability

A standard six-sided die (a 'd6') is often treated as an ideal, fair, and cubical object in probability theory. This section explores the simple mathematical odds associated with rolling a single die.

1.1 Calculating Simple Odds

The probability of rolling any specific number (n) on a fair N -sided die is given by the ratio of the number of successful outcomes (always 1) to the total number of possible outcomes.

- A standard die has a sample space size of $N = 6$.
- The chance of rolling a 4 is simply $1/6$.
- We assume the die is perfectly balanced and the roll is truly random.

The probability $P(n)$ of rolling a single number is therefore defined as:

$$P(n) = \frac{1}{N} \quad (1)$$

The Workflow

Section 1

- Create a Section
- Create a Subsection
- Create an Itemized List
- Create an Equation

```
\section{Introduction: Ideal Probability}
```

A standard six-sided die (a 'd6') is often treated as an ideal, fair, and cubical object in probability theory. This section explores the simple mathematical odds associated with rolling a single die.

```
\subsection{Calculating Simple Odds}
```

The probability of rolling any specific number (n) on a fair N -sided die is given by the ratio of the number of successful outcomes (always 1) to the total number of possible outcomes.

```
\begin{itemize}
```

```
\item A standard die has a sample space size of  $N=6$ .
```

```
\item The chance of rolling a 4 is simply  $1/6$ .
```

```
\item We assume the die is perfectly balanced and the roll is truly random.
```

```
\end{itemize}
```

The probability $P(n)$ of rolling a single number is therefore defined as:

```
\begin{equation}
```

```

$$P(n) = \frac{1}{N}$$

```

```
\end{equation}
```

1 Introduction: Ideal Probability

A standard six-sided die (a 'd6') is often treated as an ideal, fair, and cubical object in probability theory. This section explores the simple mathematical odds associated with rolling a single die.

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- We assume the die is perfectly balanced and the roll is truly random.

The probability $P(n)$ of rolling a single number is therefore defined as:

$$P(n) = \frac{1}{N} \tag{1}$$

The Workflow

Section 2

- Create a Section
- Refer a Figure
- Insert an Image

```
\section{Visualization of a Standard Die}
```

Figure `\ref{Fig:Die}` captures two standard six-sided dice mid-roll, suspended in the air. It perfectly illustrates the dynamic phase where physics meets probability, the instant before the chaotic movement results in a fixed, random outcome.

```
\begin{figure}[tbh!]  
  \centering  
  \includegraphics[width=6cm]{Dice Image.png}  
  \caption{A standard die}  
  \label{Fig:Die}  
\end{figure}
```

2 Visualization of a Standard Die

Figure 1 captures two standard six-sided dice mid-roll, suspended in the air. It perfectly illustrates the dynamic phase where physics meets probability, the instant before the chaotic movement results in a fixed, random outcome.



Figure 1: A standard die

The Workflow

Section 3

- Create a Section
- Create a Subsection
- Format Text (Subscript)
- Create an Equation

```
\subsection{Energy Transfer and Rotation}
```

The rotational kinetic energy is the energy the die possesses due to its spinning motion. This energy must dissipate against the table surface before it settles. The rotational kinetic energy (K_{rot}) is calculated as:

```
\begin{equation}
```

```
K_{rot} = \frac{1}{2} I \omega^2
```

```
\end{equation}
```

Where I is the mass moment of inertia and ω (ω) is the angular velocity in radians per second.

3 The Mechanics of Rolling

When a die is thrown, it exhibits complex dynamics, combining linear translation and angular rotation. The outcome is determined by the total kinetic energy applied and the moment of inertia, I , of the die.

3.1 Energy Transfer and Rotation

The rotational kinetic energy is the energy the die possesses due to its spinning motion. This energy must dissipate against the table surface before it settles. The rotational kinetic energy (K_{rot}) is calculated as:

$$K_{rot} = \frac{1}{2} I \omega^2 \quad (2)$$

Where I is the mass moment of inertia and ω (ω) is the angular velocity in radians per second.

| The Workflow

Section 4

- Create a Section
- Format Text (Bold)
- Create an Equation inline

```
\section{Conclusion}
```

The study of dice rolling demonstrates a blend of `\textbf{simple probability}` defined by $P(n) = \frac{1}{N}$ and complex mechanical motion governed by energy transfer. For truly random results, a die must be perfectly cubic and balanced, minimizing factors like the rotational kinetic energy.

4 Conclusion

The study of dice rolling demonstrates a blend of **simple probability** defined by $P(n) = \frac{1}{N}$ and complex mechanical motion governed by energy transfer. For truly random results, a die must be perfectly cubic and balanced, minimizing factors like the rotational kinetic energy.

The Final Challenge

Add an List of Contents

```
\begin{document}

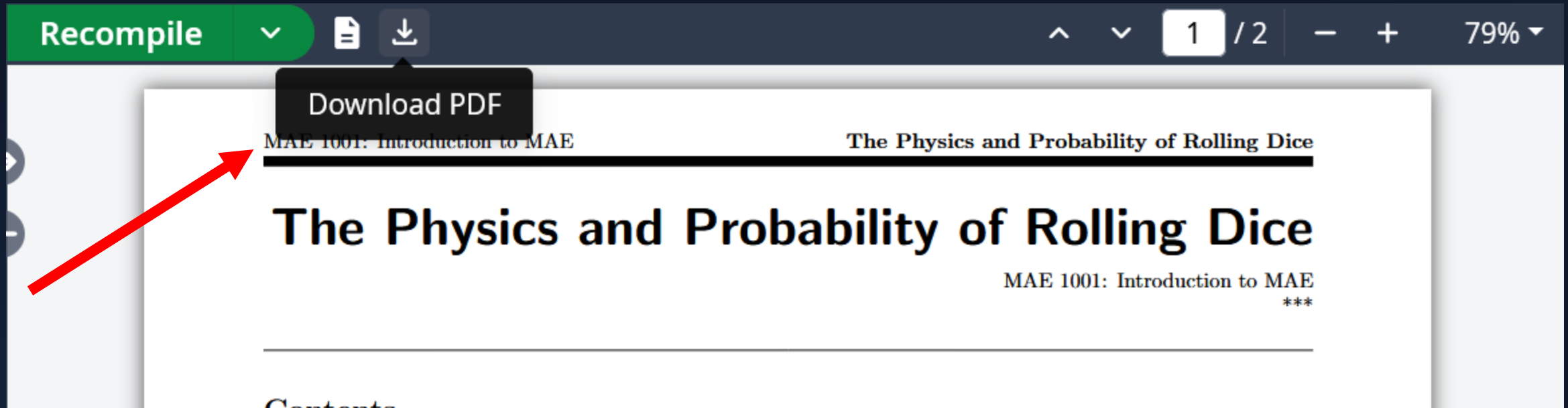
\maketitle % Generates the title, author, and date block
\tableofcontents
```

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3	The Mechanics of Rolling	2
3.1	Energy Transfer and Rotation	2
4	Conclusion	2

Assignment

Submit the final PDF file with 4 sections and Contents



The screenshot shows a Beamer presentation viewer interface. At the top, there is a dark blue header bar with a green 'Recompile' button on the left, followed by a dropdown arrow, a document icon, a download icon, and navigation controls including up/down arrows, a page indicator '1 / 2', and zoom controls '- + 79%'. Below the header, a white rectangular preview of the presentation slide is shown. A black tooltip with the text 'Download PDF' is positioned over the download icon in the header. A red arrow points from the left side of the image towards the 'Download PDF' tooltip. The slide preview itself has a white background with the text 'MAE 1001: Introduction to MAE' in the top left and 'The Physics and Probability of Rolling Dice' in the top right. A thick horizontal line separates the header from the main content area, which contains the title 'The Physics and Probability of Rolling Dice' in a large, bold font. Below the title, the text 'MAE 1001: Introduction to MAE' and '***' are visible. At the bottom of the slide, the word 'Contents' is partially visible.

Recompile

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